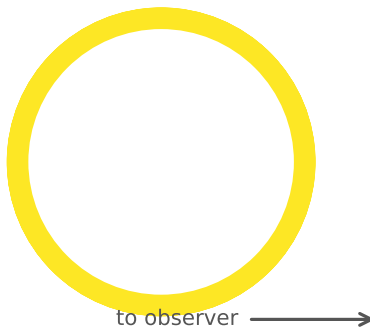
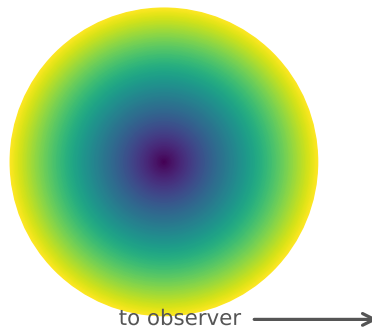


The same $1/\sqrt{v_c^2 - v^2}$ kernel in radio astronomy: removing the slow gas puts horns back into the 21 cm line

a. ring: one circular speed



b. disk: speeds all the way to zero



c. disk with a hole: sharp minimum speed

